# M3-bis2. COUPLED OSCILLATIONS

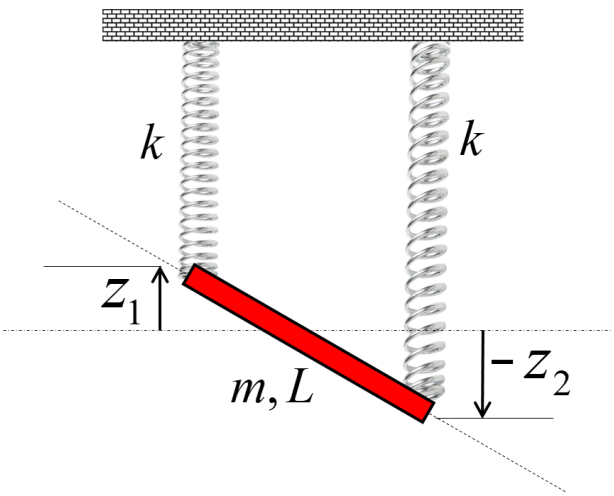
## OBJETIVES

The aim of this practice is to illustrate some of the phenomena which emerge when a rigid body is subjected to oscillations, revealing an inertial mechanical coupling. In particular, the normal modes and characteristic frequencies will be studied.

## THEORETICAL FOUNDATION

The oscillation of a rigid body attached to different springs can be really complex, but it is possible to describe it as a superposition of what is called ***normal modes***. Each normal mode does have one well-defined frequency, known as the ***characteristic frequency*** of the mode. Furthermore, it is possible to start the motion with certain initial conditions that allow us to exclusively excite one of these modes. The number of normal modes of a system is equal to its number of degrees of freedom.

Consider a homogeneous bar of mass *m* and length *L* vertically held at both ends by two springs of elastic constant *k*. This system is described by the coordinates with respect to *the equilibrium of the system* ($z_1$, $z_2$). Small oscillations are assumed.




For this system (check it), the expressions for the kinetic and potential (elastic plus gravitational) energies are:

$$T = \frac{1}{2}m\dot{z}_{CM}^2 + \frac{1}{2}I_{CM}\omega^2 = \frac{1}{8}m(\dot{z}_1 + \dot{z}_2)^2 + \frac{1}{24}m(\dot{z}_1 - \dot{z}_2)^2 \tag{1}$$

$$U = \frac{1}{2}kz_1^2 + \frac{1}{2}kz_1^2 \tag{2}$$

This system has two degrees of freedom, so it only has two normal modes. It can be proved (do so) through different methods that the frequencies of the normal modes are given by:

$$\omega_s = \sqrt{\frac{2k}{m}} \tag{3}$$

$$\omega_a = \sqrt{\frac{6k}{m}} \tag{4}$$

which are the so-called **symmetric** (slow) and **antisymmetric** (fast) modes, respectively. These are described by the normal coordinates $z_{CM}$ and $z_{12} \equiv z_1 - z_2$.

## METHODOLOGY

The experimental setup consists of 2 identical vertical springs that hold an horizontal bar (its mass is irrelevant in this practice) and they are connected to two PASCO sensors (resolution 0.03 N) which measure the force exerted by each spring. Starting from the position of equilibrium of the bar, run the `M3bis2` program on the PC desktop and press for a few seconds the ZERO button on each PASCO sensor.

Always starting from the position of equilibrium of the bar (make sure of it), record the force data corresponding to the oscillations (>10 cycles), initiated according to the following experimental situations:

a) Using the auxiliary support with the sphere-magnet system, keep one end of the bar fixed. Move the other end *slightly* in the vertical direction and release. Determine the natural frequency $\omega_o$ corresponding to the motion of a fixed bar at one end that oscillates due to the action of a spring ($k_{eff}=k$). Repeat three times. **Hint:** The bar behaves like a point mass of $m/3$.

b) To observe the symmetrical oscillation mode, move the bar *slightly* in the vertical direction without changing its orientation (pull both ends or the center of mass) and release it. Both ends should describe the same pure sinusoidal function ($z_1=z_2$). Obtain the frequency $\omega_s$ of each end and average both values. Repeat three times.

c) Using the auxiliary support with a rod, place it just below the center of mass of the bar, without varying its vertical equilibrium position. The antisymmetric mode is obtained by simultaneously displacing both ends of the bar by means of two *small*, opposite and identical vertical displacements. Alternatively, you can simultaneously hit both ends of the bar by two *small*, fast, vertical, opposite and identical pulses with the bar at rest. Both ends should describe pure but inverted sinusoidal functions ($z_1=-z_2$). Then obtain the frequency $\omega_a$ for each end and average both values. Repeat it three times.

d) To observe the superposition of modes, significantly and slowly displace one end while the other is at rest (make sure of it) and release them. Obtain the oscillation frequencies $\omega_{min}$ and $\omega_{max}$ for each end and average the values. Repeat three times.

## RESULTS

1. To determine the period of **pure** harmonic oscillations you can count the cycles in a time interval, fit the data to a sinusoidal curve or calculate the **Fast Fourier Transform (FFT)** of the signal (NOTE: FFT requires signals centered at 0, with enough cycles and what it is obtained is the frequency *f*). The frequency resolution of the FFT is the inverse of the time interval of the analyzed signal. Compare the results.

2. Complete the table below indicating uncertainties and units:

| $\omega_o$ | $\omega_s$ | $\omega_a$ |
|---|---|---|
| | | |

Should there be any numerical relation between these three frequencies? Reason it using eq. (3) and (4).

3. When motion is complex (superposition of modes), fit the data using `DataStudio` (or `Gnuplot`) to a sinusoidal series (the sum of two sine curves with a phase shift) or calculate the **Fast Fourier Transform (FFT)** (NOTE: FFT requires signals centered at 0, with enough cycles and what it is obtained is the frequency *f*). The frequency resolution of the FFT is the inverse of the time interval of the analyzed signal. Complete the following table indicating uncertainties and units:

| $\omega_{min}$ | $\omega_{max}$ |
|---|---|
| | |

Compare (relative error) $\omega_{max}$ with $\omega_a$ and $\omega_{min}$ with $\omega_s$. Reason what is observed.

**4.** In the case of a superposition of modes, since each measured force is directly proportional to the respective $z$ position, calculate $z_{CM}$ and $z_{12} \equiv z_1 - z_2$ ignoring multiplicative factors. Plot them as a function of time and obtain the frequencies. Complete the following table indicating uncertainties and units:

| $\omega_{CM}$ | $\omega_{12}$ |
|---|---|
| | |

Compare (relative error) $\omega_{12}$ with $\omega_a$ and $\omega_{CM}$ with $\omega_s$. Reason what is observed.

## BIBLIOGRAPHY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